AP CALCULUS AB

Unit 1 • Lesson 1.1 practice • Answer key

Lesson 1.1 • Introducing Calculus: Can Change Occur at an Instant?

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

For $s(t) = t^2$, which expression gives the average rate of change on $[2,\ 2+h]$, $h \neq 0$?

(A) $4+h$

(B) 4

(C) $2+h$

(D) $4h+h^2$

Solution

Answer: (A). $\dfrac{(2+h)^2-4}{h} = \dfrac{4h+h^2}{h} = 4+h.$

Question 2

NO CALCULATOR

Using the previous question, what is the instantaneous rate of change of s at $t=2$?

(A) 2

(B) 4

(C) 0

(D) It cannot be found.

Solution

Answer: (B). As $h \to 0$, $4+h \to 4$.

Question 3

NO CALCULATOR

The average velocities of an object on $[3, 3+h]$ are 6.1, 6.01, 6.001 for $h = 0.1, 0.01, 0.001$. Which is the best estimate of its velocity at $t=3$?

(A) 6.1

(B) 6.001

(C) 6

(D) 3

Solution

Answer: (C). The average velocities approach 6 as h approaches 0.

Question 4

NO CALCULATOR

What is the instantaneous rate of change of $f(x) = 3x + 2$ at $x = 5$?

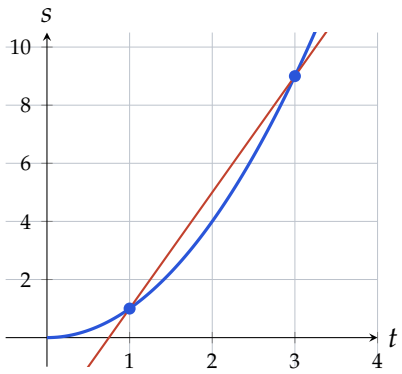
(A) 17
(B) 5
(C) 2
(D) 3

Solution
Answer: (D). A linear function has the same rate of change, its slope, at every point.

Question 5 (graph)

NO CALCULATOR

The graph of the position $s(t) = t^2$ is shown with the secant line through the points where $t = 1$ and $t = 3$. What is the slope of this secant line (the average velocity on $[1, 3]$)?



(A) 4
(B) 2
(C) 8
(D) 3

Solution
Answer: (A). $s(1) = 1$ and $s(3) = 9$, so the slope is $\dfrac{9-1}{3-1} = 4$.

Question 6 (Free response)

NO CALCULATOR

An object's position is $s(t) = t^3$ meters at time t seconds.

(a) Show that the average velocity on $[1, 1+h]$ is $3 + 3h + h^2$.
(b) Compute it for $h = 0.1$ and $h = 0.01$.
(c) Estimate the velocity of the object at $t = 1$, with units.

Solution
(a) $\dfrac{(1+h)^3 - 1}{h} = \dfrac{1 + 3h + 3h^2 + h^3 - 1}{h} = 3 + 3h + h^2$.
(b) $h = 0.1$: 3.31 m/s. $h = 0.01$: 3.0301 m/s.
(c) As $h \to 0$ the average velocity approaches 3: the velocity at $t = 1$ is 3 meters per second.